

2(a)(i)	any four from: 1 antibiotic acts as selection pressure ; 2 bacteria with antibiotic resistance (gene / allele) have selective advantage / selected for ; ora 3 bacteria with antibiotic resistance (gene / allele), survive / reproduce ; ora 4 pass on, gene / allele, (for antibiotic resistance) or increase in allele frequency (for antibiotic resistance) ; 5 <i>ref. to</i> plasmids ; 6 (horizontal gene transfer) by, transduction / transformation / conjugation ; 7 (horizontal gene transfer) gives rapid increase of resistant bacteria ;	4
2(a)(ii)	any two from: 1 drug prevents attachment of mRNA to ribosome or drug prevents attachment of tRNA to, amino acid / ribosome / mRNA or drug binds to ribosome ; 2 protein (coded for by antibiotic resistance gene) not made ; 3 so antibiotic able to work ;	2

Question	Answer	Marks
2(b)(i)	any three from: accept ora throughout <i>natural selection</i> 1 environment as selection pressure ; ora human acts as selection pressure 2 random mating ; ora humans select organisms to breed 3 takes more generations for effects to be seen / slower process / AW ; 4 organisms <u>not</u> selected for desirable phenotypes ; ora humans select features 5 can result in speciation ; 6 does <u>not</u> decrease, genetic diversity / hybrid vigor ; 7 inbreeding less likely to occur ; 8 greater heterozygosity ;	3
2(b)(ii)	1 offspring (from F1) will not be genetically similar / AW ; 2 offspring (from F1) will produce variation in next crop / will not be true-breeding plants;	2